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B.Tech / M.Tech (Integrated) DEGREE EXAMINATION, NOVEMBER 2024

Third Semester

21CSS201T - COMPUTER ORGANIZATION AND ARCHITECTURE

(For the candidates admitted from the academic year 2021-2022 to 2023-2024)

Note:

(i) **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

(ii) **Part - B & Part - C** should be answered in answer booklet.

Time: 3 hours

Max. Marks: 75

PART - A (20 x 1 = 20 Marks)
Answer ALL Questions

Marks BL CO PO

- | | | | | |
|--|---|---|---|---|
| 1. Reflected binary code is also known as | 1 | 1 | 1 | 2 |
| A) BCD Code | | | | |
| B) Binary Code | | | | |
| C) ASCII Code | | | | |
| D) Grey Code | | | | |
| 2. Which of the following is the correct hexadecimal representation for the decimal number 255 | 1 | 1 | 1 | 2 |
| A) FF | | | | |
| B) F0 | | | | |
| C) FE | | | | |
| D) F1 | | | | |
| 3. What is the 2's complement of the binary number 00001101 | 1 | 2 | 1 | 2 |
| A) 11110010 | | | | |
| B) 11110011 | | | | |
| C) 11110001 | | | | |
| D) 11101101 | | | | |
| 4. What is the gray code equivalent of the binary number 11011 | 1 | 2 | 1 | 2 |
| A) 10110 | | | | |
| B) 10001 | | | | |
| C) 11101 | | | | |
| D) 10010 | | | | |
| 5. Which of the register/s of the processor is/are connected to Memory Bus? | 1 | 1 | 2 | 2 |
| A) Personal Computer | | | | |
| B) Memory Address Register | | | | |
| C) Instruction Register | | | | |
| D) Program Counter | | | | |
| 6. Data is transferred to and from memory in groups of bits called | 1 | 1 | 2 | 2 |
| A) Words | | | | |
| B) Address | | | | |
| C) Logic gates | | | | |
| D) Bit Number | | | | |
| 7. 1 GB is equivalent to how to many bytes | 1 | 1 | 2 | 2 |
| A) 2^{10} bytes | | | | |
| B) 2^{20} bytes | | | | |
| C) 2^{50} bytes | | | | |
| D) 2^{30} bytes | | | | |
| 8. A computer has 32 MB (megabytes) of memory. How many bits are needed to address any single byte in memory | 1 | 2 | 2 | 2 |
| A) 30 bits | | | | |
| B) 25 bits | | | | |
| C) 26 bits | | | | |
| D) 20 bits | | | | |
| 9. Mention the adder circuit which can add two binary numbers and also take into account an input carry? | 1 | 2 | 3 | 2 |
| A) Half Adder | | | | |
| B) Full Adder | | | | |
| C) Ripple Adder | | | | |
| D) Look-Ahead Adder | | | | |

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10. Which algorithm is commonly used for binary multiplication in digital circuits? 1 1 3 2
 A) Booth's algorithm B) De Morgan's theorem
 C) Ripple-carry multiplication D) Carry-lookahead multiplication
11. What is the primary disadvantage of using a straight forward shift-and-add multiplier 1 2 3 2
 A) High complexity B) High speed
 C) Large area usage D) Long computation time
12. How many bits are used for the exponent in IEEE 754 standard for single-precision floating-point number representation 1 2 3 2
 A) 8 B) 11
 C) 23 D) 32
13. The stages of 3 stage pipelining are 1 1 4 2
 A) Decode, Fetch, Execute B) Execute, Fetch, Decode
 C) Fetch, Decode, Execute D) Address generation, Fetch, Execute
14. Control Hazard is otherwise called as 1 1 4 2
 A) Data Hazard B) Instruction Hazard
 C) Structural Hazard D) Information Hazard
15. A set of microinstructions for a single machine instruction is called 1 1 4 2
 A) Program B) Command
 C) Micro program D) Micro command
16. The disadvantage/s of the hardwired approach is 1 1 4 0
 A) It is less flexible B) It cannot be used for complex instructions
 C) It is costly D) Less flexible & cannot be used for complex instructions
17. Which of the following is a primary reason for using parallelism in computing 1 1 5 2
 A) To improve software compatibility B) To decrease the power consumption
 C) To increase number of processor cores for better performance D) To increase the complexity
18. In Flynn's classification, MIMD refers to 1 1 5 2
 A) Single Instruction stream, Single Data stream B) Multiple Instruction streams, Single fragment stream
 C) Single Instruction stream, Multiple Data streams D) Multiple Instruction streams, Multiple Data streams
19. Instruction level Parallelism achieves more than one instruction at a time through 1 1 5 2
 A) Dynamic scheduling B) Priority scheduling
 C) Time-slot scheduling D) Double scheduling
20. Which instruction set architecture is typically used in ARM processors 1 1 5 2
 A) Complex Instruction Set Computer B) Reduced Instruction Set Computer
 C) Very Long Instruction Word D) Explicitly Parallel Instruction Computing

PART - B (5 x 8 = 40 Marks)
 Answer ALL Questions

Marks BL CO PO

- 21 a. Describe briefly about Universal Gates? Show how to implement the basic gates using Universal Gates. 8 2 1 2

(OR)

b. Given the binary numbers 101101 and 110110, perform the following tasks

8 2 1 2

1. Add the binary numbers and provide the result.
2. Convert the binary result to its decimal equivalent.
3. Convert the decimal result to Octal and Hexadecimal

22 a. List and discuss about different addressing modes with suitable examples

8 2 2 2

(OR)

b. Describe the basic functional blocks of computer system and discuss about the functions of the Arithmetic and Logic Unit (ALU) and the Memory Unit (MU) in the basic structure of a computer.

8 2 2 2

23 a. Draw the circuit diagram of 4 bit Full adder circuit and explain its Operations.

8 3 3 2

(OR)

b. Describe the significance of Booth algorithm for performing signed multiplication with in.

8 3 3 2

24 a. Illustrate the micro-programmed control unit(MCU) with a neat diagram and how instructions will be processed in MCU.

8 2 4 2

(OR)

b. Describe the process of pipelining in computer architecture. Explain different stages of pipeline and its role in execution of instructions.

8 2 4 2

25 a. Briefly write about architecture of parallel systems and its types.

8 2 5 2

(OR)

b. Describe various memory load and store instructions available in ARM architecture.

8 2 5 2

PART - C (1 x 15 = 15 Marks)
Answer ANY ONE Question

Marks BL CO PO

26. Explain the architecture of 8086 microprocessor and briefly describe its operations.

15 2 2 2

27. Describe the architecture of ARM processors, focusing on how pipeline stages contribute to instruction execution.

15 3 5 2
